

Practice Problems -4

An object 2.50 cm high is placed 12.0 cm in front of a thin lens of focal length 3.0 cm. Calculate (a) the image distance, (b) the magnification, and (c) the nature of the image. (d) Check your answers by a graph.

A plano-convex lens is to be made of light flint glass of index 1.680. Calculate the radius of curvature necessary to give the lens a power of 4.5 D.

A double-convex lens is to be made of glass having a refractive index of 1.580. If one surface is to have twice the radius of the other and the focal length is to be +6.0 cm, find the radii.

Three thin lenses have powers +1.50, -2.80, and 3.40 D, respectively. What are all the possible powers that can be obtained with these three lenses using one, two, or three at a time in contact?

Three lenses with focal lengths of +8.40, -4.60, and +6.20 cm, respectively, are located one behind each other in this order and 2.0 cm apart. (a) If parallel light is incident on the first lens, how far behind the third lens will the light be brought to a focus? (b) Draw a scale diagram.